

CHAPTER 1

Sampling and sampling distribution

1.1. Sampling theory

1.1.1. Basic Definitions

Population: is totality of statistical information on a particular character of all the members covered by an investigation. Statistical information is 'finite' or 'infinite' according to its size. A population is said to be finite when the number of members of the population can be expressed as a definite quantity otherwise it is infinite population.

Population size: is the total number of members of the population.

Parameter: Characteristic or measure obtained from a population.

A Sample: is a portion or part of the population of interest.

Sample size: is the number of items included in the sample.

Statistic: Characteristic or measure obtained from a sample.

Sampling: The process or method of sample selection from the population.

Sampling unit: the ultimate unit to be sampled or elements of the population to be sampled

Examples: If somebody studies Socio-economic status of the households, households are the sampling unit.

If one studies performance of freshman students in certain colleges, the student is the sampling unit.

Sampling frame: is the complete list of all units in finite population.

Examples: List of households.

List of students in the registrar office.

1.1.2. The Need for Samples

There are many reasons for sampling a population.

- a) The results of a sample may be adequately estimating the value of the population parameter, thus saving time and money.
- b) It may be too time consuming to contact all members of the population.
- c) It may be impossible to check or locate all the members of the population.
- d) The cost of studying all items in the population may be prohibitive.
- e) Often testing destroys the sampled item and it cannot be returned to the population.

1.1.3. Designing and conducting a sampling study

This section describes how to select a sample. First describe how to sample from a finite population and then describe how to select a sample from an infinite population.

Finite Population

Statisticians recommend selecting a probability sample when sampling from a finite population because a probability sample allows them to make valid statistical inferences about the population. The simplest type of probability sample is one in which each sample of size n has the same probability of being selected. It is called a simple random sample.

Infinite Population

Sometimes we want to select a sample from a population, but the population is infinitely large or the elements of the population are being generated by an on-going process for which there is no limit on the number of elements that can be generated. Thus, it is not possible to develop a list of all the elements in the population. We cannot select a simple random sample because we cannot construct a frame consisting of all the elements. Statisticians recommend selecting what is called a random sample.

A random sample of size n from an infinite population is a sample selected such that the following conditions are satisfied.

- ☞ Each element selected comes from the same population.
- ☞ Each element is selected independently.

Situations involving sampling from an infinite population are usually associated with a process that operates over time. Examples include parts being manufactured on a production line, repeated experimental trials in a laboratory, transactions occurring at a bank, telephone calls arriving at a technical support center, and customers entering a retail store. In each case, the situation may be viewed as a process that generates elements from an infinite population. As long as the sampled elements are selected from the same population (i.e., must be selected at approximately the same point in time) and are selected independently (i.e., selecting of one element must not affect the next selection), the sample is considered a random sample from an infinite population.

1.1.4. Bias and Errors in Sample Survey

a) Sampling error:

Samples are used to estimate population characteristics. For example, the mean of sample is used to estimate the population mean. However, since the sample is a part or portion of the population, it is unlikely that the sample mean would be *exactly equal* to the population. Similarly, it is unlikely that the sample standard deviation would be *exactly equal* to the population standard deviation. We can therefore expect a difference between a sample *statistic* and

its corresponding population *parameter*. This difference is called **sampling error**.

Consequently, sampling error is the discrepancy between the population value and sample value. In other word it is the difference between a sample statistic and its corresponding population parameter.

Example: Consider the population of five employees at ABC Industries. Last week the output for each employee was 97, 103, 96, 99, and 105 units.

- ✓ Suppose a sample of two employees was selected and their outputs were 97 and 105. The mean of this sample is **101** $[(97 + 105)/2]$.
- ✓ Another sample of two employees resulted in outputs of 103 and 96, so the mean of this sample is **99.5**
- ✓ The mean of all the outputs (the population mean) is 100, found by

$$\mu = \frac{\sum X}{N} = \frac{97 + 103 + 96 + 99 + 105}{5} = \mathbf{100}$$

- ✓ The sampling error for the first sample is 1.0 (sample mean - $\mu = 101 - 100$) whereas the sampling error for the second sample is - 0.5 (sample mean - $\mu = 99.5 - 100$). each of these differences, 1.0 and -0.5, is the sampling error made in estimating the population mean based on the sample mean.
- ✓ Sampling error may arise due to inappropriate sampling techniques applied.

b) Non Sampling Errors

This kind of error is known as bias. Bias arises solely due to human factors and difficult to detect. They are errors due to procedure bias such as:

- ✓ Due to incorrect responses and measurement problems
- ✓ Errors at different stages in processing the data.

1.1.5. Types of Samples

There are two types of sampling.

1. Random Sampling or probability sampling.

It is a method of sampling in which all elements in the population have a pre-assigned non-zero probability to be included in to the sample. Samples are selected using the principles of probability.

It is classified into four types.

- ✓ Simple random sampling
- ✓ Systematic sampling
- ✓ Stratified random sampling
- ✓ Cluster sampling

Simple Random Sampling

It is a method of selecting items from a population such that every possible sample of specific size has an equal chance of being selected. Or all elements in

the population have the same pre-assigned non-zero probability to be included in to the sample.

Simple random sampling can be done either using the lottery method or table of random numbers. In practice the members of the sample are drawn one by one. There are two ways of drawing a simple random sample.

i) Simple Random Sampling With Replacement (SRSWR)

Simple Random Sample is said to be "with replacement", when the sample members are drawn from the population one by one, and after each drawing, the selected population unit is returned to the population before the next one is drawn. i.e., at each stage of the sampling process all the population units (including those obtained in earlier drawings) are considered for selection with equal probability.

The population remains the same before each drawing and any of the population units may appear more than once in the sample.

ii) Simple Random Sampling Without Replacement

Simple Random Sample is said to be "without replacement", when either the sample members are drawn all at a time, or drawn one by one in such a manner that after each drawing the selected unit is not returned to the population when the next one is drawn.

At each stage of the sampling process the population units already chosen are not considered for subsequent selections. The size of the population goes on diminishing as the sampling process continues. Consequently, no population unit can appear more than once in the sample.

Systematic Sampling

A complete list of all elements within the population (sampling frame) is required. The items or individuals of the population are arranged in some order. In a systematic sampling, a random starting point is selected using a simple random sampling method and then every k^{th} member of the population is selected.

The steps involved in constructing a systematic sampling scheme are as below:

Step 1: Calculate k . It is the population size is divided by the sample size.

$$N = \text{population size}, n = \text{sample size}, k = \frac{N}{n} = \text{sampling interval}.$$

Step 2: Select any number between 1 and k by using random sampling.

Suppose it is j ($1 \leq j \leq k$).

Step 3: Starting with this number select every k^{th} number until all the n units are selected. The j^{th} unit is selected at first and then $(j + k)^{\text{th}}$, $(j + 2k)^{\text{th}}$, ..., etc until the required sample size is reached.

Stratified Random Sampling

When a population can be clearly divided into groups based on some characteristic, and then stratified random sampling can be used to guarantee that each group is represented in the sample. The groups are also called **Strata**. Once the strata are defined, we can apply simple random sampling within each group or strata to collect the sample. Elements in the same strata should be more or less homogeneous while different in different strata. It is applied if the population is heterogeneous.

Some of the criteria for dividing a population into strata are: Sex (male, female); Age (under 18, 18 to 28, 29 to 39); Occupation (blue-collar, white collar, others).

Example 1: Population has 25 students of whom 15 are white and 10 black. A stratified sample of size 10 should have how many whites / blacks? Let N =population size, N_1 =blacks, N_2 =whites, n =sample size.

$N_1 / N = (10/25)*10$ or 4 blacks and $(N_2/N)*n = (15/25)*10$ or 6 we have a representative sample

Cluster Sampling

Another common type of sampling is **cluster sampling**. It is often employed to reduce the cost of sampling a population scattered over a large geographic area. In cluster sampling, a population is divided into clusters using naturally occurring geographic or other boundaries. Then, clusters are randomly selected and a sample is collected by randomly selecting from each cluster.

Clusters are formed in a way that elements within a cluster are heterogeneous, i.e. observations in each cluster should be more or less dissimilar. Cluster sampling is useful when it is difficult or costly to generate a simple random sample. For example, to estimate the average annual household income in a large city we use cluster sampling, because to use simple random sampling we need a complete list of households in the city from which to sample.

To use stratified random sampling, we would again need the list of households. A less expensive way is to let each block within the city represent a cluster. A sample of clusters could then be randomly selected, and every household within these clusters could be interviewed to find the average annual household income.

2. Non Random Sampling or Non-Probability Sampling.

The fundamental difference between non-probability sampling and probability sampling is that in non-probability sampling procedure, the selection of the sample units does not ensure a known chance to the units being selected. In other words, the units are selected without using the principle of probability.

It is a sampling technique in which the choice of individuals for a sample depends on the basis of convenience, personal choice or interest. Even though the non-probability sampling has advantages such as reduced cost, speed, and convenience in implementation, it lacks accuracy in view of the selection bias. Another negative point of the non-probability sampling is its inability to generalize results from the sample to the population. It is mandatory in inferential statistics to use only probability sampling for valid conclusions. Non-probability sampling is suitable for pilot studies and exploratory research. It is classified into three:

- ✓ Judgment sampling.
- ✓ Convenience sampling
- ✓ Quota Sampling.

Judgment (Purposive) Sampling

A sample which is selected on the basis of individual judgment of the sampler, is called a *Purposive Sample*. There is no special technique for selecting a purposive sample, but the sampler picks out a typical or representative sample according to his own judgment. Consequently, there is much scope for bias and the degree of accuracy of the estimates is not known.

Purposive sampling may be useful when the sample is small; but as the sample size increases the estimates become unreliable due to accumulation of bias. The advantage of purposive sampling is that whereas a random sample may vary widely from the average, a purposive sample will not.

Convenience Sampling

In this method, the decision maker selects a sample from the population in a manner that is relatively easy and convenient. The researcher samples whatever units come most readily to hand. In this case, probability is not used in the sampling at all.

Quota Sampling

In this method, the decision maker requires the sample to contain a certain number of items with a given characteristic. Many political polls are, in part,

quota sampling. In simple terms, quota sampling is stratified random sampling without probability principle being applied to the selection of the sample units.

Example: Suppose in an opinion study, you want both men and women to participate. You know that in the population category of interest, 65% are men and 35 % are women. If your sample size is fixed at 200, you will have a quota of 130 men and 70 women.

3.2. SAMPLING DISTRIBUTION

3.2.1. Definitions

Sampling distribution is defined as the sampling distribution of the mean; the sampling distribution of the mean is the probability distribution of the sample mean. If we organize the arithmetic means of all possible sample of size n from a given population size N into a probability distribution, we will get the sampling distribution of the mean.

In other words, it is a probability distribution consisting of a list of all possible sample mean of a given sample size selected from the population and the probability of occurrence associated with each sample mean.

Sampling distribution of the mean is a probability distribution of all possible sample means of a given sample size.

3.2.2. Sampling Distribution of the Mean

Steps for the construction of sampling distribution of the mean

1. From a finite population of size N , randomly draw all possible samples of size n .
2. Calculate the mean for each sample.
3. Summarize the mean obtained in step 2 in terms of frequency distribution or relative frequency distribution.

Example: Consider a population with size of four ($N = 4$) and it includes variables 2, 3, 5 and 6. If samples with the size of two are to be taken without replacement from the population

- a) Compute the population mean
- b) How many possible sample arrangements will be there?
- c) Identify the possible samples.
- d) Compute the sample means.
- e) Develop the sampling distribution of the mean.
- f) Determine the mean of the sampling distribution of the sample mean.

Solution

a) $\mu = \frac{2 + 3 + 5 + 6}{4} = 4$

b) ${}^nC_x = {}^4C_2 = \frac{4!}{2!(4-2)!} = 6$ samples

c) $n_1 = (2, 3), n_2 = (2, 5), n_3 = (2, 6), n_4 = (3, 5), n_5 = (3, 6),$ and $n_6 = (5, 6)$

d) $\bar{x}$ for $n_1 = \frac{2+3}{2} = 2.5$, $\bar{x}$ for $n_2 = \frac{2+5}{2} = 3.5$, $\bar{x}$ for $n_3 = \frac{2+6}{2} = 4$, $\bar{x}$ for $n_4 = \frac{3+5}{2} = 4$, $\bar{x}$ for $n_5 = \frac{3+6}{2} = 4.5$, $\bar{x}$ for $n_6 = \frac{5+6}{2} = 5.5$

e)

$\bar{x}$	2.5	3.5	4	4.5	5.5
$f(\bar{x})$	1	1	2	1	1
$P(\bar{x})$	1/6= 0.1667	1/6= 0.1667	2/6=0.333	1/6=0.1667	1/6=0.1667

f) The mean of the sampling distribution of the sample mean is obtained by summing the various sample means and dividing the sum by the number of the samples. The mean of all the sample means is usually written $\mu_{\bar{x}}$.

$$\mu_{\bar{x}} = \text{Sum of all sample means} = \frac{2.5 + 3.5 + 4 + 4 + 4.5 + 5.5}{6} = \frac{24}{6} = 4$$

In summary, this example illustrates important relationships between the population distribution and the sampling distribution of the sample mean.

1. The mean of the sample means is exactly equal to the population mean.
2. The dispersion of the sampling distribution of sample means is narrower than the population distribution. This is because the spread in the distribution of the sample mean ranges from 2.5 to 5.5, while the population values vary from 2 up to 6.
3. The shape of the sampling distribution of the sample mean and the shape of the frequency distribution of the population values are different. The distribution of the sample mean tends to be more bell-shaped and to approximate the normal probability distribution.

Properties of the Sampling Distribution of $\bar{x}$

1. The mean of the sample means is exactly equal to the population mean.

$$\mu = \mu_{\bar{x}}$$

2. In general if sampling is Infinite or,
Finite with less than or equal to 5% of the population sample size

Then the standard deviation of the distribution of the sample mean is equal to the population standard deviation divided by the square root of the sample size.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad \text{Otherwise,} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\left(\frac{N-n}{N-1}\right)}$$

Note that as we increase the size of the sample, the spread of the distribution of the sample mean becomes smaller.

Example: The standard deviation of annual salary for the population of 2500 managers is $\sigma = 4000$. In this case, the population is finite, with $N = 2500$. With a sample size of 30, compute the standard deviation of the sample mean.

Solution: we have $n/N = 30/2500 = .012$. Because the sample size is less than 5% of the population size, we can ignore the finite population correction factor and use the infinite population factor.

$$\sigma_{\bar{x}} = \frac{4000}{\sqrt{30}} = \underline{\underline{30.3}}$$

Take the same example above and if the sample size is 1000 instead of 30, compute the standard deviation of the sample mean.

Solution: we have $n/N = 1000/2500 = 0.4$. Because the sample size is more than 5% of the population size, we can use the finite population correction factor.

$$\begin{aligned} \sigma_{\bar{x}} &= \frac{4000}{\sqrt{1000}} \sqrt{\left(\frac{2500-1000}{2500-1}\right)} \\ &= \underline{\underline{98}} \end{aligned}$$

Form of the Sampling Distribution of $\bar{X}$

The preceding results concerning the expected value and standard deviation for the sampling distribution of $\bar{X}$ are applicable for any population. The final step in identifying the characteristics of the sampling distribution of $\bar{X}$ is to determine the form or shape of the sampling distribution. We will consider two cases: (1) The population has a normal distribution; and (2) the population does not have a normal distribution.

Population has a normal distribution. In many situations it is reasonable to assume that the population from which we are selecting a random sample has a normal, or nearly normal, distribution. When the population has a normal distribution, the sampling distribution of $\bar{X}$ is normally distributed for any sample size.

Population does not have a normal distribution. When the population from which we are selecting a random sample does not have a normal distribution, the **central limit theorem** is helpful in identifying the shape of the sampling distribution of $\bar{X}$. A statement of the central limit theorem as it applies to the sampling distribution of $\bar{X}$ follows.